MPCs, MPEs, and Multipliers:
A Trilemma for New Keynesian Models

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Introduction

- **The Micro-Moment:** Macroeconomics increasingly uses microeconomic data to discipline and calibrate structural models.
- **The Core Tension:** Dynamic models must simultaneously be consistent with:
    1. *Micro consumption behavior:* How households respond to income shocks.
    2. *Micro labor supply:* How households adjust earnings in response to wealth.
    3. *Macro policy multipliers:* General equilibrium effects of government spending.
- **The Problem:** Frictionless labor supply models face an **Trilemma** between these moments.
- **The Solution:** Pushing households off their short-run labor supply curves via **Sticky Wages** resolves the tension, matching all moments simultaneously.

Three Key Empirical Facts

Fact 1: High MPCs

Average Marginal Propensities to Consume are high in the data:

- **Quarterly MPC:** $\approx \mathbf{0.25}$
- **Annual MPC:** $\approx \mathbf{0.50}$

Fact 2: Low MPEs

Average Marginal Propensities to Earn are extremely low:

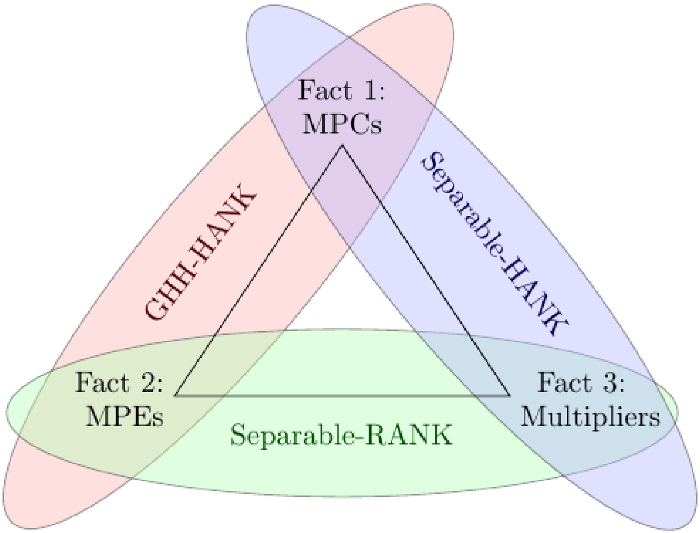
- MPE estimated annual to be $\approx \mathbf{0.01}$ by Cesarini et al. 2017).
- Near zero across micro studies.

Fact 3: Modest Multipliers

Fiscal multipliers are modest, even under accommodative monetary policy:

- Empirical consensus under constant real r or peg lies between $\mathbf{1.0}$ and $\mathbf{1.5}$.
- Multipliers do not explode to extreme levels (> 3).

Figure 1: The New Keynesian Trilemma



The New Keynesian Trilemma

Model Class	Fact 1 (MPC ≈ 0.25)	Fact 2 (MPE ≈ 0)	Fact 3 (Multiplier ≈ 1–1.5)
RANK	**Fail** (≈ 0.05)	**Pass** (≈ 0.00)	**Pass** (1.0 under separable)
HANK (Separable)	**Pass** (≈ 0.25 by calib.)	**Fail** (≈ 0.20 wealth effect)	**Pass** (≈ 1.20)
HANK (GHH)	**Pass** (≈ 0.25 by calib.)	**Pass** (0.00 by design)	**Fail** (> 3.0 or 5.5)

Table: The Performance of Frictionless Labor Supply Models

- **Separable preferences** hit the multiplier but fail micro labor supply (wealth effect makes households work less).
- **GHH preferences** shut down the wealth effect (MPE = 0) but force an explosive macro multiplier.

Separable vs. GHH Preferences

Separable Preferences

$$\mathcal{U}(c,n) = \frac{c^{1-\sigma}}{1-\sigma} - \psi \frac{n^{1+1/\nu}}{1+1/\nu}$$

- **Cross-Derivative:** $\mathcal{U}_{cn} = 0$.
- **Labor Supply FOC:**

$$(1-\tau^w) w c^{-\sigma} = \psi n^{1/\nu}$$

$$\implies n = \left[\frac{(1-\tau^w) w c^{-\sigma}}{\psi}\right]^{\nu}$$

- **Intuition:** Transfer $T \uparrow \implies c \uparrow \implies c^{-\sigma} \downarrow \implies n \downarrow$. Wealth effect makes households work less.

GHH Preferences

$$\mathcal{U}(c,n) = \frac{1}{1-\sigma}\left(c - \psi \frac{n^{1+1/\nu}}{1+1/\nu}\right)^{1-\sigma}$$

- **Cross-Derivative:** $\mathcal{U}_{cn} \neq 0$ (non-separable).
- **Labor Supply FOC:**

$$(1-\tau^w) w = \psi n^{1/\nu}$$

$$\implies n = \left[\frac{(1-\tau^w) w}{\psi}\right]^{\nu}$$

- **Intuition:** $MRS_{c,n}$ depends only on labor. Labor supply is independent of consumption or wealth. No wealth effect on labor (MPE = 0).

The MPC-MPE Relationship

- **Budget & FOCs:** Face effective wage $\tilde{w} \equiv (1 - \tau^w)w$. Household optimization:

$$c + a' = \tilde{w}n + T + (1+r)a \quad \Longrightarrow \quad U_c(c,n) = \lambda, \quad U_n(c,n) = -\lambda\tilde{w}$$

- **Complementarity Index (CI):** Measures consumption-labor complementarity holding λ fixed:

$$\text{CI} \equiv \frac{U_{cn}}{U_{cc}}\frac{1}{\tilde{w}} = \frac{\frac{\partial c(\lambda,\tilde{w})}{\partial \tilde{w}}}{\tilde{w}\frac{\partial n(\lambda,\tilde{w})}{\partial \tilde{w}}} \quad \Longrightarrow \quad \text{CI} = 0 \text{ (Separable)}, \quad \text{CI} = 1 \text{ (GHH)}$$

Proposition 1: The Micro Link

Under frictionless labor supply, individual MPC and MPE are related by:

$$\text{MPE} = (1 - \text{CI}) \cdot \text{MPC} \cdot \frac{\tilde{w}n}{c} \cdot \frac{\text{Frisch}}{\text{EIS}}$$

where Frisch $\equiv \frac{\partial \log n}{\partial \log \tilde{w}}$ and EIS $\equiv -\frac{\partial \log c}{\partial \log \lambda}$ are idiosyncratic elasticities.

Since MPC is high (≈ 0.25), to get low MPE (≈ 0), we **must** have CI ≈ 1 (GHH).

CI and the Multiplier

- Consider a Representative Agent (RANK) model under constant real interest rates.

Proposition 2: General Equilibrium Multiplier

Under a constant real interest rate, the fiscal multiplier is given by:

$$\frac{dY_t}{dG_s} = \frac{1}{1-(1-\tau)\text{CI}} \cdot \mathbf{1}_{s=t}$$

where $\tau < 1$ is the steady-state labor wedge.

Two Extreme Cases:

- **Separable Preferences (CI $= 0$):** Multiplier = 1.0 (Woodford 2011). No consumption-labor feedback loop. Matches Fact 3.
- **GHH Preferences (CI $= 1$):** Multiplier = $\frac{1}{\tau}$. For typical calibrations where the labor wedge is small ($\tau \approx 0.2$, so $1-\tau \approx 0.8$), the multiplier is $\frac{1}{\tau} \approx 5.0$ Fails Fact 3.

Quantitative Frictionless HANK Setup

Can a heterogeneous-agent structure break this tight analytical micro-macro link?

"GHH-plus" Preferences

$$U(c,n) = \frac{1}{1-\sigma}\left(c - \phi\alpha\frac{n^{1+\nu}}{1+\nu}\right)^{1-\sigma} - \phi(1-\alpha)\frac{n^{1+\nu}}{1+\nu}$$

where $\alpha \in [0,1]$ regulates complementarity: $\alpha = 0 \implies$ Separable, $\alpha = 1 \implies$ GHH.

- **Calibration Strategy:** For each $\alpha \in [0,1]$, the household asset/income distribution is recalibrated to hit a quarterly average MPC of **0.25** (Fact 1).

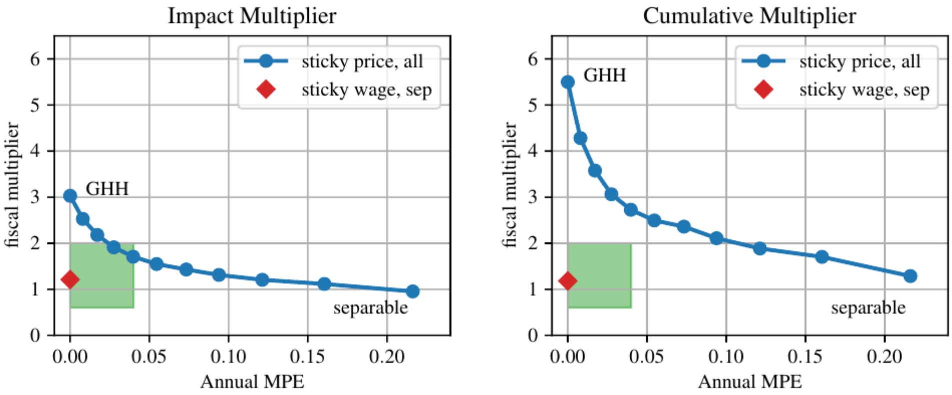
Frictionless Trilemma: Quantitative Results

Preferences	Parameter α	Complementarity (CI)	Annual MPE	Constant-r Multiplier
Separable	0.0	0.00	0.19 (Too High)	1.21 (Plausible)
Intermediate	0.5	0.50	0.09 (Too High)	2.07 (Too High)
GHH	1.0	1.00	0.00 (Correct)	5.50 (Explosive)

Table: Quantitative Moments in Frictionless HANK (Calibrated to MPC = 0.25)

- **The Frictionless Trilemma:** There is no parameter $\alpha \in [0, 1]$ that can simultaneously deliver low MPE (Fact 2) and moderate multipliers (Fact 3).
- **Robustness:** This trilemma is robust to varying the Frisch elasticity and risk aversion.

Figure 2: The Multiplier-MPE Trade-off



The Solution: Sticky Wages & Demand-Determined Labor

- **Wage Stickiness (Erceg et al. 2000):** Labor unions set nominal wages subject to adjustment costs; households supply demanded hours: $n_{it} = N_t$.
- **Zero Wealth Effect:** A windfall transfer has **zero effect** on short-run labor supply $\implies$ MPE $= \mathbf{0}$ by construction (Fact 2 matched).
- **Flexible Prices:** Constant markup fixes the real wage: $w_t = Z \frac{e_p - 1}{e_p}$.
- **Wage Phillips Curve:**

$$\log(1 + \pi_t^w) = \kappa_w \left(N_t U_t^n - \frac{e_w - 1}{e_w} (1 - \tau_t^w) w_t N_t U_t^c \right) + \beta \log(1 + \pi_{t+1}^w)$$

where $U_t^n \equiv \frac{1}{N_t} \int n_{it} U_n di$ (avg marginal disutility of work) and $U_t^c \equiv \int e_{it} U_c di$ (productivity-weighted avg marginal utility of consumption).

Quantitative Results with Sticky Wages

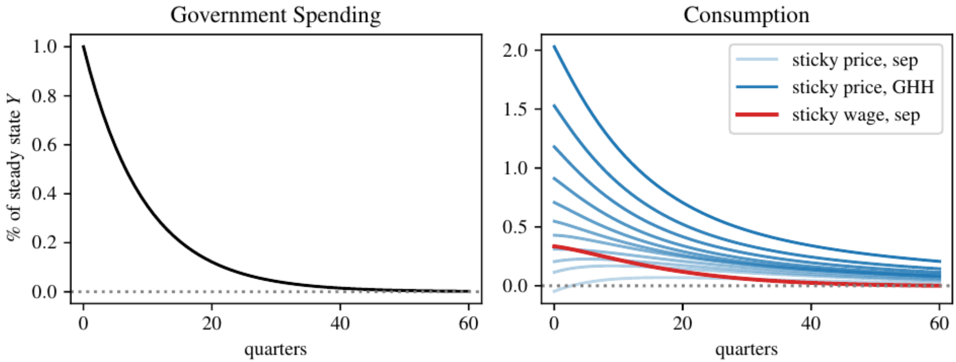
- Authors calibrate the sticky-wage HANK model with **separable preferences** ($\alpha = 0$, CI $= 0$) to hit a quarterly average MPC of 0.25.

Model Version	Quarterly MPC	Annual MPE	Constant-r Multiplier
Frictionless Separable	0.25	0.19	1.21
Frictionless GHH	0.25	0.00	5.50
Sticky Wage Separable	$\mathbf{0.25}$	$\mathbf{0.00}$	$\mathbf{1.21}$ (Impact) / $\mathbf{1.18}$ (Cum.)

Table: Comparing Sticky-Wage and Frictionless Models

- **Resolution of the Trilemma:**
  - *Wage stickiness* absorbs the micro wealth effect on labor $\implies$ MPE $= 0$.
  - *Separable preferences* keep consumption-labor complementarity low (CI $= 0$) $\implies$ prevents the general equilibrium multiplier from exploding.

Figure 3: Private Consumption Response



Conclusion

- **The New Keynesian Trilemma:** Frictionless models face an trilemma: they cannot simultaneously match high MPCs, low MPEs, and moderate fiscal multipliers.
- **The Problem:** Frictionless labor supply forces a trade-off: either accept high wealth effects (high MPE) or high complementarity (high multipliers).
- **The Solution:** Sticky wages. By shifting short-run adjustments away from labor supply curves, sticky wages decouple micro wealth effects from macro multipliers.
- **Takeaway:** Nominal wage stickiness is an essential element in HANK models to align disaggregated micro behavior with aggregate policy transmission.